Retail Sales Performance 2025

العربية

Overview

A retail sales analysis built entirely in **Power BI**, with all data cleaning and preparation done in **Power Query**.

Business Problem

Where do revenue and profit come from, and how do sales change through the year?

Questions

- Which product categories are the most profitable?
- How do sales change month by month?
- Which cities and sales channels perform best?

Dataset

- **Synthetic dataset** generated for practice (not real sales)
- `sales_orders.csv` : 3,245 raw order rows, one year (2025)
- `products.csv` : 30 products in 6 categories, with cost
- Intentional data quality issues were included to practice cleaning

Tools

- Power BI
- Power Query

Data Cleaning (Power Query)

Issue	Fix

Duplicate orders	Removed on `Order_ID`
Dates in two formats	Converted to the Date type
City names written in different ways (for example riyadh, Riyad)	Trim, proper case, then Replace Values with "Match entire cell contents"
Extra spaces in payment method	Trim
Text "SAR" inside prices	Removed, then converted to Decimal Number
Discount as 10%, as 0.1, or blank	Converted to a single decimal format, blank set to 0
Missing quantities (~1.5%)	Rows removed, because revenue cannot be calculated
Negative quantities (returns)	Flagged in a new column `Order_Type` (Return or Sale), returns excluded from sales
Missing customer ID (~2%)	Kept, because it does not affect sales or profit analysis

Data Preparation

- Merged orders with products (Left Outer join on `Product_ID`) to bring in category, product name, and cost
- `Revenue = Quantity x Unit_Price x (1 - Discount)`
- `Profit = Revenue - Quantity x Cost`

Key Findings

- Total revenue is about 1.27M SAR and total profit is about 593K SAR, across about 3,145 sales orders.
- Revenue has two peaks: March (about 170K) and November (about 132K).
- Home is the most profitable category (about 162K), followed by Clothing (about 103K). Grocery is the lowest (about 25K).
- Riyadh leads revenue by a clear margin (about 393K), then Jeddah (about 258K) and Dammam (about 128K).
- Online is the largest sales channel (about 46% of revenue).

Limitations

- The data is synthetic, so findings describe this dataset only.
- The profit margin (about 47%) is higher than typical for real retail because costs were simulated.
- Seasonal peaks come from the data generation, not from real events.

Files

- `sales_orders.csv` and `products.csv` : datasets
- `retail_sales_dashboard.pbix` : Power BI report (cleaning steps are visible in Power Query)
- `power_query/01_cleaning_steps.m` : cleaning code from the Advanced Editor
- `images/` : dashboard and Applied Steps screenshots

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